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Lab work

Lab 1: File System (fs)

1. Create a file named data.txt with some text

```
const fs = require("fs");

// Create file
fs.writeFileSync("data.txt", "Hello, this is initial text.");
console.log("File created successfully!");
```

2. Read the file contents

```
const fs = require("fs");

fs.readFile("data.txt", "utf8", (err, data) => {
  if (err) throw err;
  console.log("File Contents:", data);
});
```

3. Append new text to the file

```
const fs = require("fs");

fs.appendFile("data.txt", "\nThis is appended text.", (err) => {
  if (err) throw err;
  console.log("Text appended successfully!");
});
```

4. Delete the file

```
const fs = require("fs");

fs.unlink("data.txt", (err) => {
  if (err) throw err;
  console.log("File deleted successfully!");
});
```

Lab 2: Operating System (os)

```
const os = require("os");
```

// 1. Platform & Architecture

```
console.log("Platform:", os.platform());
console.log("Architecture:", os.arch());
```

// 2. Total & Free memory

```
console.log("Total Memory:", os.totalmem());
console.log("Free Memory:", os.freemem());
```

// 3. Home directory

```
console.log("Home Directory:", os.homedir());
```

Lab 3: HTTP (http)

1 & 2. Create simple HTTP server on port 3000

```
const http = require("http");

const server = http.createServer((req, res) => {
  res.write("Hello World");
  res.end();
});
```

```
server.listen(3000, () => {  
  console.log("Server running at http://localhost:3000/");  
});
```

3. Test in Browser

Open:

<http://localhost:3000>

Lab 4: Path (path)

```
const path = require("path");  
  
// 1. Directory name  
console.log("Directory Name:", __dirname);  
  
// 2. Base name  
console.log("Base Name:", path.basename(__filename));  
  
// 3. Join paths  
console.log("Joined Path:", path.join("/users", "docs"));
```

Lab 5: Events (events)

```
const EventEmitter = require("events");  
const emitter = new EventEmitter();  
  
// 2. Listener  
emitter.on("greet", (msg) => {  
  console.log("Greet Event Received:", msg);  
});  
  
// 3. Emit event
```

```
emitter.emit("greet", "Hello from the event emitter!");
```

Lab 6: Util (util)

1. Util.format

```
const util = require("util");
```

```
let formatted = util.format("My name is %s and I am %d years old",  
"Ragini", 20);  
console.log(formatted);
```

2. Util.promisify

```
const fs = require("fs");  
const util = require("util");
```

```
const readFilePromise = util.promisify(fs.readFile);
```

```
readFilePromise("data.txt", "utf8")  
  .then(data => console.log("Promise File Data:", data))  
  .catch(err => console.log(err));
```

Lab 7: Streams (stream)

1–3 Copy file using Streams

```
const fs = require("fs");
```

```
// Create read stream
```

```
const readStream = fs.createReadStream("source.txt");
```

```
// Create write stream
```

```
const writeStream = fs.createWriteStream("destination.txt");
```

```
// Pipe data flow
```

```
readStream.pipe(writeStream);
```

```
console.log("File copying using streams... (non-blocking)");
```